Analytical Chemistry Cumulative Examination
December 5, 2014

1. A recent paper entitled " New Structural Insights into Mechanically Interlinked Polymers Revealed by Ion Mobility Mass Spectrometry" was published in *J. Amer. Chem. Soc.*, **134**, pp. 9193-9198 (2012). Please describe the specific type of polymer the research concerns. What are the main observations that ion mobility mass spectrometry revealed? This study involved use of a traveling wave ion mobility system. What is traveling wave ion mobility? How do the cross sections derived from those measurements differ from non-traveling wave ion mobility cross sections and can the two be related?

2. A review paper by Benesch, et. al. addresses the investigation of protein complexes in the gas phase (*Chem. Rev.* **107**, pp. 3544-3567 (2007)). Please explain how ion mobility mass spectrometry is used in this context. Name one such protein complex that was studied by this technique and explain what information was derived from those studies.

3. An interesting study of fragmentation and isomerization due to field heating in traveling wave ion mobility mass spectrometry was reported in a paper published in the August, 2014 issue of *J. Am. Soc. Mass Spectrom.*, (**25,** pp. 1384-1393) by Morsa, et. al.. Why is heating observed? What were some of the analytes employed in this study? How much heating was seen? What is the importance of this effect for large biomolecules? Why?

Biochemistry Cumulative Examination
12/05/2014

In your structures drawing for this exam you may use Fischer, Haworth, a plain old skeletal structure, or whatever style you prefer, and switch back and forth as you wish, but in all your drawings the stereochemistry must be clear.

1. What sugar(s) make(s) up sucrose? 10 points

2. Draw the structure of glucose. 11 points. Is the common form D or L? 3 points. Which did you draw? 1 point.

3. Draw the structure of N-acetylglucosamine. 15 points

4. Where are some places one commonly finds N-acetylglucosamine? 15 points

5. What sugar is found in DNA? Name it. 3 points. Draw the structure. 12 points

6. What are the two simplest (smallest) sugars typically commonly found in cells (albeit usually phosphorylated)? Name and draw each in unphosphorylated form. 20 points

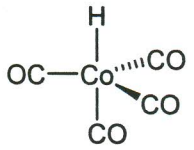
7. What is the name we use for the carbon in a sugar that often has two oxygens bonded to it? 10 points

Inorganic Cumulative Exam

December 5, 2014

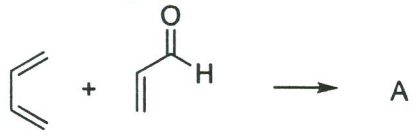
Provide a complete mechanism for the catalytic reaction shown below. For each step give the oxidation state of the metal catalyst. For each step indicate what compounds are entering and leaving the catalytic cycle.

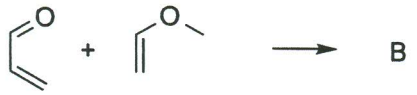
catalyst :



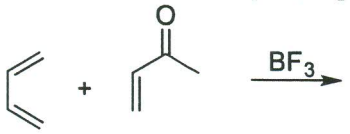
Organic Cume 12-5-2014

1. According to the FMO (Frontier Molecular Orbital) theory, draw the HOMO (highest occupied molecular orbital) and the LUMO (lowest occupied molecular orbital) of the following Diels-Alder reactions that have the most important interactions. Give the structure of the products **A** and **B**.

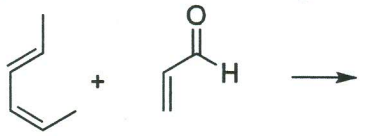
1) 

2) 

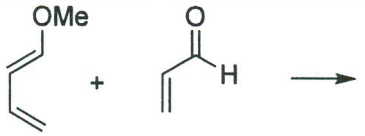
2. Lewis acids are known to accelerate the Diels-Alder reactions. Using the following reaction as an example, explain the rate acceleration based on the FMO theory.



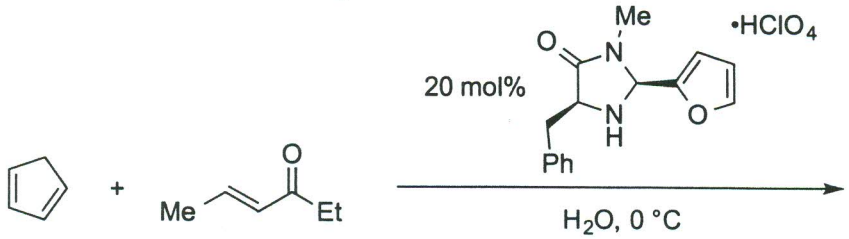
3. The endo selectivity is often observed in the Diels-Alder reactions. Draw the endo product of the following reaction. Use the FMO theory to explain the endo selectivity.



4. The regioselectivity may arise when unsymmetrical dienes and dienophiles are used in the Diels-Alder reactions. Give the major regioisomer of the following reaction (you can ignore the endo/exo selectivity and the facial selectivity with respect to the diene and the dienophile). Use resonance structures to explain the regioselectivity.



5. Give the major product with stereochemistry of the following Diels-Alder reaction. Rationalize the sense of asymmetric induction for this reaction.



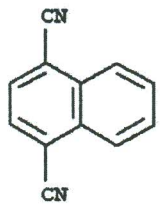
Physical Chemistry Cumulative Exam – Dec 5th 2014

The following paper was previously announced:

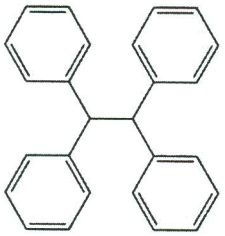
Kinetics of C–C and C–H Bond Cleavage in Phenyl Alkane Radical Cations Generated by Photoinduced Electron Transfer. Douglas Cyr and Paritosh Das. Journal of Physical Chemistry A, 2014, 118 (47), pp 11155–11167

Necessary figures and information from the paper is given in the appendix to help you answer the questions

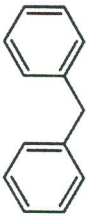
In addition, you may find the following structures helpful:



1,4-dicyanonaphthalene, DCN



1,1,2,2-tetraphenylethane, TPE



diphenylmethane, DPME

1) The paper describes the use of nanosecond laser flash photolysis, transient absorption spectroscopy and time-resolved fluorescence. Describe your understanding of these techniques and how they are employed in this study. You do not need to give technical details – just focus on the conceptual aspects.

(35 points)

2) Explain how the authors used figures 1 and 2 (see appendix for these figures) to form one of the main conclusions to their paper; that the C-C bond cleavage contributes about 70% and deprotonation contributes about 30% to the fragmentation of the 1,1,2,2-tetraphenylethane (TPE) radical cation (see relevant processes in appendix). You do not need to go through the math – just describe their process.

Hint: Equation (9) from the paper is

$$\phi_{C-C} = \frac{k_5}{k_5 + k_6} = 2.6\frac{\Delta A_{max,Ph_2CH^+}}{\Delta A_{max,DCN^{-\bullet}}}\frac{\varepsilon_{max,DCN^{-\bullet}}}{\varepsilon_{max,Ph_2CH^+}} \qquad (9)$$

(40 points)

3) In figure 1 of the paper (see appendix), the authors showed a time-resolved fluorescence decay curve at 550 nm. This was measured by using a second nanosecond laser pulse of 337.1 nm on the sample several microseconds (~10 μs) after the first (308 nm) nanosecond laser pulse.

a) Which species did the authors assign this fluorescence in figure 1 to?

b) The authors supported their assignment of the fluorescent species by a subsequent experiment in which either 1,1,2,2-tetraphenylethane (TPE) or diphenylmethane (DPME) were exposed to di-tert-butyl peroxide in benzene (BZ-DTBP), which generated tert-butoxy radicals upon exposure to the 308 nm laser, resulting in hydrogen abstraction of TPE according to:

$$t\text{-Bu—O—O—Bu-}t + h\nu \rightarrow 2t\text{-BuO}^{\bullet}$$

$$t\text{-BuO}^{\bullet} + \text{Ph}_2\text{CH—CHPh}_2 \rightarrow t\text{-BuOH} + \text{Ph}_2\text{C}^{\bullet}\text{—CHPh}_2$$

hydrogen abstraction of DMPE (in place of TPE) proceeds via the same mechanism

The result from this experiment is given in figure 3 (see appendix for the figure). Explain how the data from figure 3 supports their assignment of the fluorescence in figure 1?

Hint: pay attention to the scale bars and the relative signal:noise ratios in the insets of figure 3. You can ignore the (light blue) Acetone/TPE curve in this figure (Acetone is just an alternative to BZ-DTBP for abstracting hydrogen from TPE).

c) How does this fluorescence data further support their conclusion that C-C bond cleavage is a significant component of the decay of the TPE radical cation?

(25 Points)

Appendix: Figures and information from the paper

for

Physical Chemistry Cumulative Exam – Dec 5th 2014

The relevant processes described in the paper for a representative donor (1,1,2,2-tetraphenylethane, TPE) and acceptor (1,4-dicyanonaphthalene, DCN) are:

$$^{1}DCN^{*} + TPE \rightarrow {}^{1}(DCN^{-\bullet}/TPE^{+\bullet})$$

(Formation of radical ion pair) (1)

$$^{1}(DCN^{-\bullet}/TPE^{+\bullet}) \rightarrow DCN + TPE$$

(Back electron transfer) (2)

$$^{1}(DCN^{-\bullet}/TPE^{+\bullet}) \rightarrow {}^{3}DCN^{*} + TPE$$

(ET-assisted intersystem crossing) (3)

$$^{1}(DCN^{-\bullet}/TPE^{+\bullet}) \rightarrow {}^{2}DCN^{-\bullet} + {}^{2}TPE^{+\bullet}$$

(Ion separation) (4)

$$^{2}TPE^{+\bullet} \rightarrow {}^{2}Ph_2CH^{\bullet} + Ph_2CH^{+} \quad \text{(C—C bond cleavage)}$$

(5)

$$^{2}TPE^{+\bullet} \rightarrow {}^{2}Ph_2C^{\bullet}\text{—}CHPh_2 + H^{+} \quad \text{(Deprotonation)}$$

(6)

$$^{2}TPE^{+\bullet} + {}^{2}DCN^{-\bullet} \rightarrow DCN + TPE$$

(Diffusional back electron transfer) (7)

Figures 1 and 2 in the paper presented results on DCN/TPE:

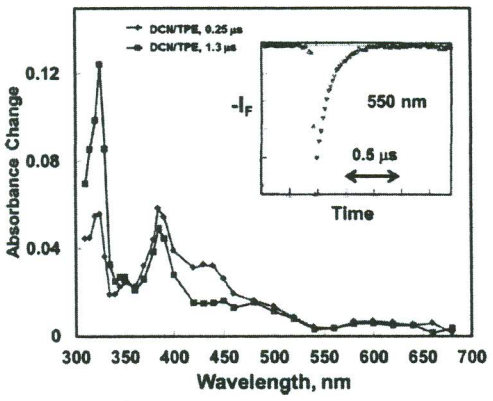


Figure 1. Transient absorption spectra observed at 0.25 and 1.3 μs following 308 nm laser pulse excitation of DCN in the presence of 20 mM TPE in acetonitrile. (Inset) Kinetic trace of decay of fluorescence resulting from subsequent 337.1 nm laser pulse excitation of the transient absorbance produced by 308 nm laser pulse excitation of DCN in the presence of 25 mM TPE in acetonitrile.

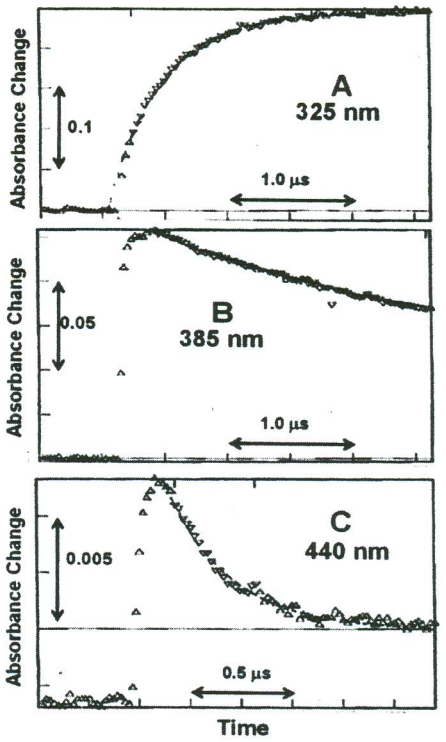


Figure 2. Representative kinetic traces of transient absorption at (A) 325, (B) 385, and (C) 440 nm following 308 nm laser pulse excitation of DCN in the presence of 20 mM TPE in acetonitrile.

According to the text in the manuscript, the peaks in the transient absorption spectra of figure 1 were assigned as follows:

325-330 nm: partially to the diphenylmethyl radical, $Ph_2CH^{\bullet}$, and partially to $Ph_2C^{\bullet}—CHPh_2$

385-390 nm: the solvated radical anion of DCN following dissociation of the ion pair $DCN^{-\bullet}/TPE^{+\bullet}$

440 nm: the diphenylmethyl carbocation, Ph_2CH^{+}

Figure 3 in the paper presented transient absorption and fluorescence results on 1,1,2,2-tetraphenylethane (TPE) or diphenylmethane (DPME) exposed to di-tert-butyl peroxide in benzene (BZ-DTBP):

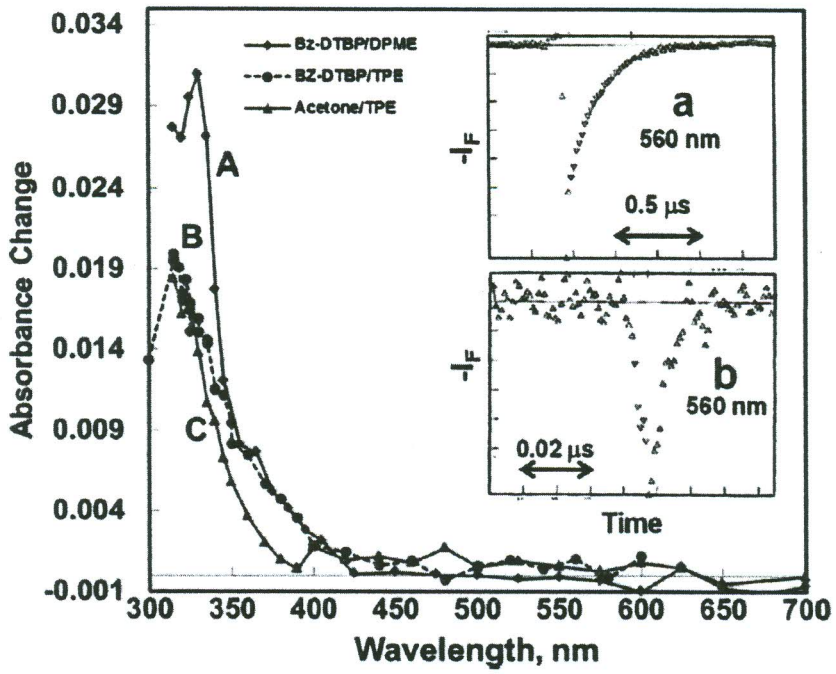


Figure 3. Transient absorption spectra observed upon 308 nm laser pulse excitation of (A) 10 mM DPME solution in benzene:DTBP mixture (4:1, v/v), (B) 10 mM TPE solution in benzene:DTBP mixture (4:1, v/v), and (C) 10 mM TPE in acetone:acetonitrile mixture (1:9, v/v). Times at which the spectra were recorded were 1, 1.2, and 1.2 μs, respectively, following the laser pulse. (Insets) (a) Kinetic trace of fluorescence at 560 nm resulting from 337.1 nm laser pulse excitation of the transient absorbance produced by 308 nm laser pulse excitation 10 mM DPME solution in benzene:DTBP mixture; (b) kinetic trace of emission at 560 nm resulting from 337.1 nm laser pulse excitation of the transient absorbance produced by 308 nm laser pulse excitation 10 mM TPE solution in benzene:DTBP mixture.